

MemeSpeak: Making Image Memes Blind-Accessible through Emotional Speech Augmentation

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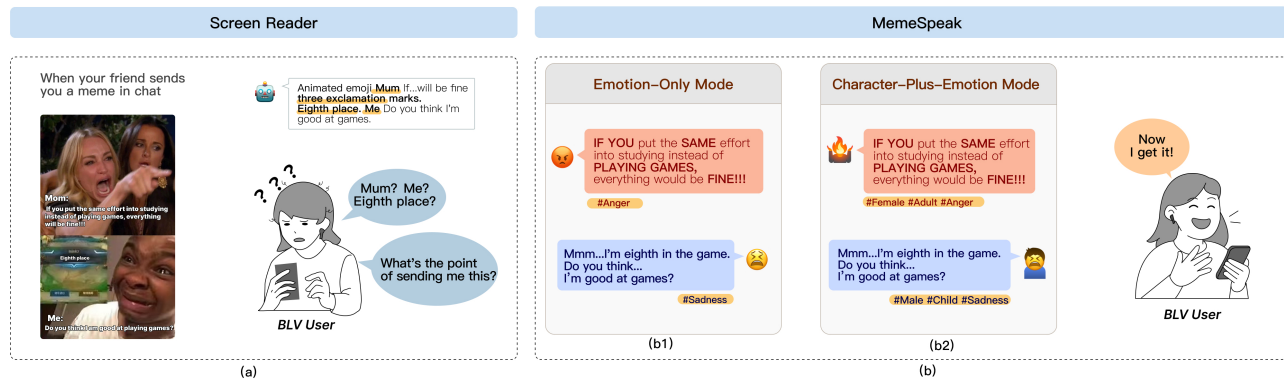


Figure 1: (a) Screen readers provide basic descriptions for image memes but fail to capture their emotional nuances. Consequently, blind and low vision (BLV) users may feel confused, excluded, or disengaged from meme-based conversations. (b) MEMESPEAK improves meme accessibility by generating emotionally expressive audio through two *emotional speech augmentation* strategies: (b1) Using emotional tones (e.g., sadness, anger) to convey affective content. (b2) Combining both emotional tone and character-specific voice variation to reflect multi-character interactions. Both strategies enable BLV users to understand, interpret, and emotionally engage with meme content more effectively.

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Abstract

Image memes—combining humorous or satirical pictures with over-laying text—are a powerful medium for emotional expression. However, their visual nature makes them largely inaccessible to blind

and low vision (BLV) users. We present MEMESPEAK, a system that generates emotionally expressive audio narrations of memes through *emotional speech augmentation*. We developed two emotion-conveying strategies: emotion-only and character-plus-emotion, and evaluated them against standard screen-reader-based descriptions with 12 BLV participants. Results showed that both strategies significantly improved emotional resonance, eliciting stronger emotional fluctuations and better understanding of the sender's intent. Qualitative feedback revealed that the character-plus-emotion strategy enhanced immersion and clarity, particularly for complex or multi-character memes. We further highlight implications for conveying visual humor in blind-accessible formats.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools**; *Empirical studies in accessibility*.

Keywords

Meme, emotion, empathy, Blind, low vision, speech, audio, large language model

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1 Introduction and Related Work

Image memes are static pictures featuring humorous or ironic text, often employing hyperbole, substitution, or subversion to engage viewers through visual and cultural references [10, 14, 18]. These memes often reflect real-world events and spread rapidly online [3]. Although screen readers can provide basic image descriptions for memes (see Figure 1 (a)), they fail to convey their emotional tone, character interactions, and cultural contexts. Consequently, blind and low vision (BLV) users are often excluded from the emotional and social depth embedded in these visual media [6].

Prior work has taken initial steps toward making memes more accessible [7–9, 15, 27], primarily by using automated methods [8, 11] and crowdsourced efforts [21, 24, 27] to generate descriptive text. However, these efforts have not fully explored the expressive potential of audio features—such as tone of voice—to convey emotion. Other studies have examined conveying information through non-visual modalities such as haptics [12, 13, 16, 20, 26], but these approaches primarily rely on specialized hardware and are not compatible with widely used assistive technologies such as screen readers. As a result, a critical gap remains in understanding how to leverage audio features beyond spoken words to effectively convey the emotional dimensions of memes to BLV users.

To address this gap, we conducted a formative study with six BLV users to understand their needs and challenges in understanding image memes. Our findings revealed a clear disconnect between standard meme descriptions and their emotional impact. Participants expressed a strong desire for more **expressive, context-rich audio** that conveyed not just the content, but also the emotional

tone and the sender's intent (e.g., to express sorrow). These insights underscored the need for emotion-aware strategies to make memes more accessible.

Building on these findings, we developed MEMESPEAK, a system that transforms memes into emotionally expressive audio through **emotional speech augmentation**. The system implements two emotion-conveyance strategies: *emotion-only*, which focuses on conveying emotions using voice tones, and *character-plus-emotion*, which combines emotional tone with character-specific voice variation. A user study with 12 BLV participants demonstrated that both strategies significantly improved emotional comprehension, immersion, and recognition of the sender's intent compared to baseline image descriptions provided by screen readers. Participants described the emotional speech augmentation as making memes more expressive and engaging. These results highlight the promise of rich audio augmentation in making visual media more emotionally resonant and accessible for BLV users.

2 Formative Study

To understand the needs and challenges that BLV users face when engaging with memes, we conducted semi-structured interviews with six BLV participants (P1–P6, summarized in Table 1). Participants were recruited through an online community. During each session, participants shared their experiences and preferences related to meme consumption and demonstrated how they accessed memes using their current methods. Each interview lasted approximately 20 minutes. We analyzed the interview transcripts using an open coding approach [2], followed by iterative synthesis to extract recurring themes. In the following sections, we present key findings in two aspects: (1) challenges faced by BLV users in understanding memes, and (2) their expectations for engaging with image memes.

2.1 Challenges in Interpreting Memes

During the study, participants reported two primary challenges in interpreting image memes.

First, current accessibility tools provide limited emotional and contextual information about memes. While screen readers can deliver basic visual descriptions, they often miss critical elements such as *emotional tone*, *character interactions*, and *cultural context*—all essential for interpreting humor or sarcasm. As a result, emotional nuance in memes is particularly difficult to grasp. For example, P6 shared that while they could identify “a cat with a big mouth”, they were still left wondering about “the cat's emotion”. This lack of depth led participants to avoid meme-heavy conversations (three times) or disregard memes entirely (two times).

Second, interpreting complex memes requires additional mental and social effort. To ensure key information is not missed, participants described relying on AI tools (five times) or reaching out to sighted friends (three times). This verification can be both time-consuming and socially burdensome. In fast-paced or informal chat settings, these barriers often lead to feelings of exclusion. As P2 explained, “If the other person knows my vision condition and sends this kind of meme without any explanation or description, I will feel uncomfortable.” These experiences highlight a deeper marginalization from emotionally rich, socially embedded communication that relies heavily on shared visual cues.

2.2 Expectations for Engaging with Memes

To better interpret and empathize with memes, participants shared two common expectations.

First, all participants expressed a strong preference for emotionally expressive formats—such as human-like audio—over purely factual descriptions. As P6 noted, “*Most AI just use factual tones to explain humor in memes, but they don’t express feeling. Explaining just kills a meme.*” Rather than simply being told what is visually present, participants wanted to experience the mood and tone conveyed by the meme.

Second, four participants also shared a broader preference for audio-based format, such as audiobooks or dubbed stories, highlighting the role of voice, tone, and sound effects in communicating emotion. As P1 noted, “*Voice helps me feel more emotion*”, while P2 added, “*It makes the context and characters more real.*”

2.3 Summary of Key Findings

Overall, our formative study revealed two primary challenges BLV users face when interpreting memes: (1) limited access to emotional and contextual information, and (2) the additional mental and social effort required to understand complex memes. To better engage with memes, participants expressed a strong preference for emotionally expressive formats—particularly human-like audio. Based on these insights, we designed MEMESPEAK.

3 MEMESPEAK

We present MEMESPEAK, a system that transforms an image meme into an emotionally expressive audio through *emotional speech augmentation*. In the following sections, we first introduce two alternative augmentation strategies and then describe the end-to-end pipeline that realises them.

3.1 Two Strategies for Emotional Speech Augmentation

MEMESPEAK supports two complementary modes: (1) *Emotion-Only* — a single narrator voice whose prosody (pitch, rate, intensity) is modulated to convey the emotion of each panel (e.g., sadness, anger); and (2) *Character-Plus-Emotion* — the same prosodic manipulation as above, but with an additional timbre shift whenever the speaker changes, yielding character-specific voices that communicate both persona and emotion.

The two modes are implemented by prompting the text-to-speech (TTS) model with distinct instruction templates: the *Emotion-Only* prompt specifies the target emotion alone, whereas the *Character-Plus-Emotion* prompt additionally names the speaking character and any relevant demographic or species cues.

3.2 Emotional Speech Generation Pipeline

The system operates in two stages (Fig. 2): (A) *meme analysis*, which extracts structured semantic and affective cues from the image; (B) *emotional speech generation*, which maps those cues to speech whose timbre and prosody mirror the intended mood. The resulting audio enables BLV audiences to perceive the emotional nuance of the original meme.

(A) **Meme Analysis.** We employ GPT-4o to encode each meme into a JSON structure that records

- the number and reading order of panels;
- identity attributes of each depicted character (e.g., profession, species);
- the coarse emotion of every character, inferred from facial expression and posture;
- verbatim text in each panel, partitioned into dialogue and narration.

(B) **Emotional Speech Generation.** Using the extracted attributes, the system invokes the OpenAI TTS API¹. For *Character-Plus-Emotion*, the prompt passes the full empathy specification, including character attributes; for *Emotion-Only*, character details are omitted. Because the current TTS model can occasionally omit or slur words, the system requests up to three candidate audio clips per utterance and selects the first that contains every word from the input.

4 Evaluation Study

We conducted a within-subjects study to evaluate whether MEMESPEAK improves BLV users’ ability to empathize with the emotional content of memes. Three system conditions were compared:

- **Baseline (C1)** – a screen-reader style description that reads only the textual content of the meme;
- **Emotion-Only (C2)** – C1 augmented with prosodic cues that convey emotions in memes;
- **Character-Plus-Emotion (C3)** – C2 further enriched with character-specific timbre changes.

Participants. We recruited twelve participants (five female, seven male; $M_{\text{age}} = 25.2$, $SD = 2.9$; P7 - P18 in Table 1). All participants have substantial online experience and basic meme literacy.

Memos. Four widely circulated memes were selected to provide diversity in content and tone (Appendix ??). The order of the three conditions was counter-balanced with a Latin-square design [4] to control for order effects.

Procedure. After hearing all the meme audio in each condition, participants filled out a seven-point Likert questionnaire that measured emotional resonance, empathy, and intent recognition. The items were adapted from prior work on empathy and emotional communication [1, 5, 23, 25]; the full questionnaire is shown in the appendix. Once all trials were complete, participants took part in a semi-structured interview and provided detailed feedback for each condition.

Analysis. All sessions were audio-recorded, transcribed verbatim, and analysed using thematic analysis [19]. The themes reported below address the effectiveness of the different augmentation strategies.

4.1 Quantitative Results

Emotional speech augmentation significantly boosts BLV users’ empathy for memes, and both augmented strategies (C2 and C3) clearly outperform the baseline. However, the two augmented modes do not differ statistically from each other. As shown in Fig. 3, a Friedman test [17] on 48 observations (12

¹<https://platform.openai.com/docs/guides/text-to-speech>

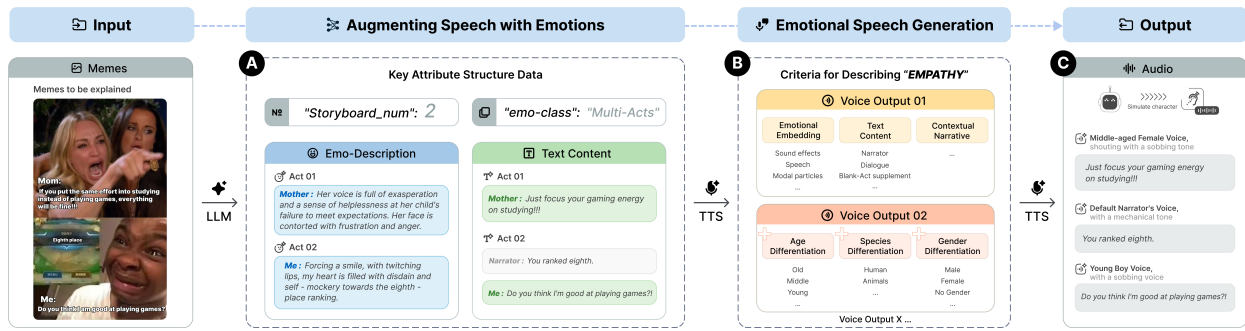


Figure 2: MEMESPEAK pipeline. (A) The meme is decomposed into panels and analysed for visual, textual, and affective attributes. (B) The system then generates emotional speech based on the extracted attributes.

participants $\times$ 4 memes) revealed significant differences among C1, C2, and C3 on all five questionnaire items ($p < .01$). Medium effect sizes were observed for Q2 ($W = .224$), Q4 ($W = .300$), and Q5 ($W = .232$). Pairwise Wilcoxon signed-rank tests [22] confirmed that both C2 and C3 surpassed C1 on emotion comprehension ($Z = -4.98$), emotional arousal ($Z = -5.07$), immersion ($Z = -4.81$), sender-intent recognition ($Z = -5.95$), and willingness to share memes ($Z = -5.48$), all $p < .01$. No Wilcoxon comparison between C2 and C3 reached significance.

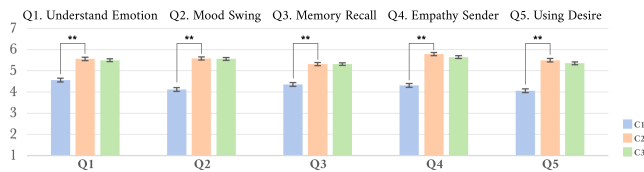


Figure 3: User ratings of the three conditions (C1, C2, and C3) across five evaluation dimensions. Ratings were collected using a 7-point Likert scale (1 = Strongly Disagree, 7 = Strongly Agree, 4 = Neutral). Error bars represent standard errors. Significant differences between conditions are marked with asterisks (** denotes $p < .01$).

4.2 Qualitative Results

(1) Potential of Emotional Speech Augmentation. All participants reported that emotional speech augmentation deepened their empathic engagement with memes. They felt that emotional speech anthropomorphized the content and captured their attention: P12 and P14 would “pay more attention to the meme,” P9 said it “felt like listening to a joke,” and P10 and P15 highlighted how onomatopoeia clarified the meme’s intent. Five participants said emotional speech helped them sense the sender’s emotion—P11 noted, “The cooing sound of the pigeon lets me feel the sender’s mood,” which made social interactions easier (P10) and avoided feeling “disconnected” (P13). For four participants, the audio also evoked personal memories (P8, P9, P14), creating emotional resonance (P13) and increasing their willingness to use memes for self-expression (P15). However, participants expressed several concerns: missing contextual explanations (P9, P14) and doubts about accuracy (P10,

P13). P9 suggested adding a brief gloss—e.g., “With the phrase ‘stand someone up,’ the joke is instantly clear.” Additionally, P11 and P12 said they trusted VoiceOver more for literal content.

(2) Role of Emotional Tones for Improving Empathy. Ten of twelve participants believed that emotional prosody helped them empathize more deeply. They described C1 as “as flat as a robot” (P11, P15), whereas C2 was “so lively I almost wanted to stand up” (P13); one-third of the group laughed aloud while listening to C2. Yet some participants found exaggerated emotion distracting (P13, P14) or unnecessary for comprehension (P11, P12). As P10 noted, “After years of mechanical TTS, it is hard to adjust to emotional reading, especially when it’s negative.”

(3) Role of Character-Specific Tones for Improving Empathy. Nine participants felt that character-specific voices improved understanding and empathy, particularly for non-human figures or multi-speaker dialogues. As P10 noted, “In C3 it sounds like two people talking; in C2 it’s the same person.” P13 and P15 noted that character-specific tones help them form a mental image of each speaker, enhancing immersion—“I can feel the mother-child interaction” (P8)—and helping them relate the meme to real life (P12). Some participants, however, noted that character-specific voices could increase cognitive load (P10, P11). P9 concluded that its usefulness “depends on the scenario.”

5 Conclusion and Future Work

We present MEMESPEAK, a system that enhances meme accessibility for BLV users through *emotional speech augmentation*. We developed two strategies—emotion-only and character-plus-emotion—and evaluated them against standard screen-reader descriptions with 12 BLV participants. Both strategies significantly improved emotional resonance, deepened empathy, and improved recognition of the sender’s intent. Qualitative feedback highlighted that the character-plus-emotion approach enhanced immersion and clarity, especially for complex or multi-character memes. Our findings suggest key design implications for conveying visual humor in blind-accessible formats.

We also identify several directions for future work. First, to better support diverse user needs, we plan to expand the voice library and enable adjustable speech parameters such as speed and

emotional intensity. Second, to improve generalizability, we aim to evaluate the system with a broader and more diverse set of memes, including culturally nuanced and visually complex instances. Third, we envision extending MEMESPEAK into a creator tool, allowing meme authors to generate emotional speech directly, fostering more inclusive content creation from the source.

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A Supplementary Materials

ID	Age	Gender	Visual Condition	Screen Reader	Social Media Usage	Where to Encounter Memes
P1	21	M	Totally blind	VoiceOver	5 - 6 hours per day	Wechat, Redbook
P2	24	M	Totally blind	Talkback	6 - 7h	Wechat, Tiktok
P3	31	M	Totally blind	TianTan	9 - 10h	Wechat, Tiktok, Twitter
P4	22	M	Totally blind	VoiceOver	3 - 4h	Wechat, Tiktok
P5	27	F	Legally blind	VoiceOver	2 - 3h	Wechat, Tiktok
P6	27	F	Totally blind	VoiceOver	4 - 5h	Wechat
P7	21	M	Totally blind	VoiceOver	5 - 6h	Wechat, Redbook
P8	24	M	Totally blind	Talkback	6 - 7h	Wechat, Tiktok
P9	31	M	Totally blind	TianTan	9 - 10h	Wechat, Tiktok, Twitter
P10	22	M	Totally blind	VoiceOver	3 - 4h	Wechat, Tiktok
P11	26	M	Totally blind	VoiceOver	3 - 5h	Wechat
P12	27	M	Totally blind	TianTan	7 - 8h	Wechat
P13	25	F	Legally blind	VoiceOver	2 - 3h	Wechat
P14	22	F	Totally blind	VoiceOver	2 - 3h	Wechat
P15	27	F	Legally blind	VoiceOver	2 - 3h	Wechat, Tiktok
P16	22	M	Totally blind	VoiceOver	3 - 4h	Wechat
P17	28	F	Legally blind	TianTan	3 - 8h	Wechat, Tiktok
P18	27	F	Totally blind	VoiceOver	4 - 5h	Wechat

Table 1: Participants' demographics. P1-P6 participated in the formative study. P7-P18 participated in the evaluation study.

- Q1 I can understand the emotions that memes are trying to convey.
- Q2 When listening to this audio, my emotions fluctuated (such as feeling happy, sad, angry, etc.).
- Q3 I can empathize with the content described in the memes or recall similar past experiences.
- Q4 I can put myself in the sender's position and understand their mood and intentions.
- Q5 This audio inspires my desire to express myself, and I would like to use this memes in similar situations in the future.

Table 2: Questionnaires used in the evaluation study. All questions used a seven-point Likert scale.



Figure 4: Four memes used in the evaluation study.